

3-Week Executive ML Curriculum: Fraud Detection

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Overview

Project: Credit card fraud/anomaly detection, taught end-to-end: frame the business problem, build a supervised baseline, build a true unsupervised anomaly detector, compare the two, then package a demo.

Dataset: Kaggle's Credit Card Fraud Detection dataset (284,807 transactions, 492 frauds, ~0.17% positive rate). Features V1-V28 are PCA-anonymized; Time and Amount are raw. The severe imbalance is the point, not a flaw: it forces real conversations about accuracy vs. precision/recall, and about what a false positive costs a real customer.

Format: 15 sessions, weekdays, 1-2 hours each, one-on-one.

Scope philosophy: this is not a tour of every ML topic. Days 1-2 are pure theory, covering only what the project needs before touching data: framing a business problem as a measurable AI task, why imbalance breaks accuracy, precision/recall/F1/ROC-AUC vs. PR-AUC, and enough distance/PCA intuition to make anomaly detection make sense later. Everything after that is taught just-in-time, inside the project, in the order the project needs it. Python fundamentals, deep learning, computer vision, NLP, and generative AI are deliberately out of scope.

Week 1: Theory, then first contact with the data

Day	Focus	What gets covered
1	Theory: framing the problem	AI vs. classical rules vs. ML; what makes fraud detection a measurable AI task; supervised vs. unsupervised framing; why this project needs both

Day	Focus	What gets covered
2	Theory: statistics for imbalanced problems	Why accuracy is meaningless at 0.17% positive rate; confusion matrix; precision, recall, F1; ROC-AUC vs. PR-AUC and when each lies to you; enough distance/PCA intuition to preview anomaly detection
3	Project kickoff: EDA	Load the Kaggle dataset, examine class balance, distributions of Amount and Time, correlation structure of V1-V28; write down the business framing in the exec's own words (cost of a missed fraud vs. cost of a false alarm)
4	Data preparation	Train/test split done correctly for imbalanced data (stratified split); scaling Amount and Time (V1-V28 are already PCA-transformed, don't re-scale them blindly, discuss why); decide whether to resample (SMOTE, undersampling) or leave imbalance in and adjust thresholds instead, and why that choice matters
5	Supervised baseline	Train logistic regression as a first baseline; look at raw output before applying any metric judgment; sets up Week 2's evaluation deep-dive

By end of Week 1, the exec has: the business framing, a clean train/test split, and a first (unjudged) model. No evaluation conclusions are drawn yet, that's deliberate, it's Week 2's first act.

Week 2: Evaluate properly, then go unsupervised

Day	Focus	What gets covered
6	Better supervised model	Random forest (or gradient boosting) against the logistic regression baseline; feature importance, what V1-V28 being anonymized costs you when explaining results to stakeholders
7	Evaluation, done right	Confusion matrix at different thresholds; precision/recall/F1 computed by hand once, then from sklearn; ROC-AUC vs. PR-AUC on this exact imbalanced dataset; threshold tuning tied to a business cost framing (cost of a missed fraud vs. cost of blocking a real customer)
8	Unsupervised framing	Pretend the fraud label doesn't exist: clustering intuition, distance-based outlier reasoning, why this maps naturally to the PCA-transformed features; connects back to Day 2's distance/PCA theory
9	Isolation Forest	Build an unsupervised anomaly detector (Isolation Forest, optionally a second method like Local Outlier Factor for contrast); score every transaction without touching the label
10	Supervised vs. unsupervised, head to head	Reveal the true labels and compare: does the unsupervised model catch what the supervised one caught; where do they disagree and why; discuss the real-world case for each (unsupervised wins when you have no labels yet, e.g. a brand-new fraud pattern)

By end of Week 2, the exec has two working models, a real understanding of why accuracy is the wrong headline metric here, and a working thesis on when you'd deploy which approach.

Week 3: Responsible AI, packaging, and the finish

Day	Focus	What gets covered
11	Responsible AI, applied	Fairness and explainability framed around this exact model: what does a false positive cost a real customer (a declined card, a frozen account); a short model card documenting what the model does, its limitations, and where it should not be trusted unsupervised
12	Packaging	Wrap the chosen final model behind a small API (FastAPI or Flask): input a transaction, output a fraud score and a flag
13	Demo build	Turn the API into something the exec can click through, a minimal interface where they enter or pick a transaction and see the model reason about it in real time
14	Documentation and presentation prep	Finalize the model card; prepare a short presentation: problem, approach, results, limitations, and what you'd need to do differently for production (real-time scoring, drift monitoring, human review queue for borderline cases)
15	Final presentation	The exec presents the project as if to their own leadership: the business framing, the two modeling approaches, the metrics that matter, the demo, and honest next steps

The capstone deliverable is the working demo plus the model card and presentation, not a notebook. That's the standard this scope was built to hit.

Resources

- [Credit Card Fraud Detection dataset \(Kaggle\)](#) — the dataset this curriculum is built around
- Core stack: Python, pandas, scikit-learn (Logistic Regression, Random Forest, Isolation Forest), matplotlib/seaborn for EDA

- Packaging: FastAPI or Flask for the demo API

Alternative if you want raw, non-anonymized features: an insurance-claims or transaction-level fraud dataset instead of the PCA-anonymized Kaggle set trades cleanliness for realism, the exec can reason about actual variables instead of V1-V28, but expect messier data prep in Week 1.